Meaning as Relational Rewiring: Quantifying Contextual Recoupling of Latent Semantics in Visual Symbolic Systems

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Abstract

Visual symbolic systems embody ambiguity, polysemy, and context-dependent meaning, qualities that static language-model embeddings flatten into averaged points. The Lakota Shape Kit, a traditional storytelling tool, exemplifies this richness. Each symbol juxtaposes many concurrent readings whose prominence shifts with narrative context. Working from Lakota relational epistemology (*Mitákuye Oyás'iŋ*, the principle that nothing exists in isolation and that meaning is a thing's position in a living web of relations), we develop a framework that represents each symbol as a context-weighted distribution over semantic descriptors and then asks not merely *which* descriptors a context foregrounds, but *how a context recouples the relations among them*. Our central instrument is the Δ coupling matrix, the signed change in the pairwise inner-product (Gram) structure of a curated lexicon under a context-shift operator. Δ localizes contextualization to the lexical relations that strengthen or weaken, providing a grounded, interpretable readout of how narrative framing rewires a symbolic field. Its dominant component is a first-order, relevance-gated reweighting of the encoder's own contextualization, re-expressed in the community-curated coordinate system, and we demonstrate it across several settings. We read a narrative as a trajectory through coupling space, use the recoupling margin to locate where a context leaves the field's meaning genuinely ambiguous, recover content-bearing context similarity once a single activation-magnitude artifact is removed, and, because the readout is modality-agnostic, probe the field's ambiguity with sound as well as text. Demonstrated with Lakota symbols and developed in collaboration with a knowledge holder under the Abundant Intelligences program, the framework advances context-sensitive meaning modeling for computational linguistics, cross-cultural knowledge representation, and creative AI.

1 The Lakota Shape Kit

Language can be understood fundamentally as a relational web, where each word derives its meaning through differences and positional relationships with all other elements in a linguistic system Stawarska [2020], Wood and Bernasconi [1988]. Words seldom possess static meanings; rather, their interpretations shift dynamically according to context, usage, and narrative positioning. This inherent polysemy and context-dependence is particularly pronounced in visual symbolic languages, systems where symbols encapsulate broad, multilayered semantic fields [Saint-Martin, 1990].

39th Conference on Neural Information Processing Systems (NeurIPS 2025) Creative AI Track.

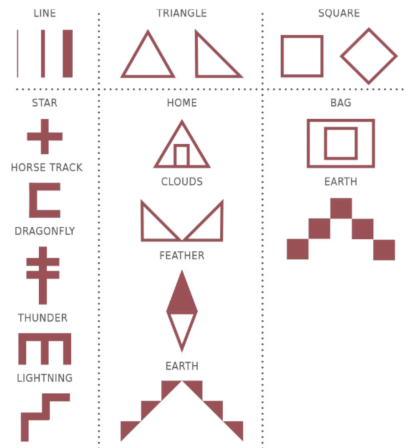

Figure 1: Lakota Shape Kit, a visual storytelling tool rooted in Lakota traditions (from [Red Wing, 2016]).

The Lakota Shape Kit Red Wing [2016] exemplifies a visual symbolic system characterized by high degrees of polysemy, interpretive openness, and nonlinear combinatoriality. Traditionally, it provides a nuanced framework for narrating dreams [Kite, 2023], inherently designed to preserve and explore multiple concurrent meanings and storylines. Each Lakota symbol, such as the dragonfly or the house, can evoke diverse and sometimes divergent meanings depending on its cultural and situational context, with the relative prominence of each meaning shifting fluidly according to the narrative surroundings.

This work is grounded in Lakota relational epistemology, expressed in the principle *Mitákuye Oyás'iŋ*, "all my relations." Meaning, in this view, is not a property an object carries in isolation but the position it occupies in a living web of relations among descriptors, contexts, and interpreters. This view constitutes the core inspiration and motivation for the proposed method. It tells us *what to look at*. If meaning is constituted by relations, then the right object of study under context is not the displacement of a symbol's vector but the reconfiguration of the relations among its facets. The framework below is a formal expression of that commitment, developed in collaboration with a Lakota knowledge holder within the Abundant Intelligences program, which bring forth AI systems built from Indigenous knowledge systems[Lewis et al., 2020].

Modern language embeddings are, in fact, remarkably good at *encoding* ambiguity. A trained model packs many senses of a word into a single vector, and contextualized encoders adjust that vector fluidly as the surrounding text changes Haber and Poesio [2024], Loureiro et al. [2021]. The difficulty is not that models cannot represent polysemy, but that this latent ambiguity is neither *legible* nor *relational*. The senses are entangled in an opaque coordinate system rather than organized by the categories of a particular symbolic tradition; meanings underrepresented in training data may be faint or absent (a limitation of data more than of architecture); and, most importantly for a relational epistemology, the encoder exposes the contextualized *vector* of a symbol but not how a context reshapes the web of *relations* among a curated field's facets, which is the level at which meaning here actually lives Ferrari et al. [2018]. The gap, then, is a gap in *instruments*. We lack ways to read how a model internally represents, and contextually reshapes, a community-curated symbolic system.

This paper addresses that gap. Large language models already shift a symbol's meaning fluidly with context (this is, in effect, what their attention layers do), but that internal machinery is rarely made inspectable at the level of a set of symbolic anchors. We therefore treat a curated lexicon as a controlled probe into a language model's latent space. Each symbol is represented as a distribution over textual descriptors, and we measure the signed change in the pairwise coupling among those descriptors under context, turning the model's internal reconfiguration into a legible, per-relation readout. We call this **quantifying how context recouples the latent semantics** of the field.

Our contribution is threefold: (1) we cast a community-curated symbolic system as a controlled probe into a language model's latent space, making its internal, context-dependent representation of that system inspectable rather than treating context-sensitivity as a black box; (2) we introduce the Δ coupling matrix, a structural probe that decomposes a context's effect into the lexical relations it strengthens or weakens; and (3) we demonstrate the readout across narrative, ambiguity, and similarity settings, and show it is modality-agnostic, probing the field's ambiguity through sound as well as through text. Ultimately this research aims to inform AI methodologies developed in active collaboration with Indigenous research projects and grounded in their knowledge systems.

2 The Inherent Ambiguity of Meaning

Ambiguity occurs when a signal supports multiple valid interpretations even when well-formed. In language, polysemy (where a single form is linked to multiple related senses) is increasingly regarded as the norm rather than the exception. Empirical and theoretical work shows that ambiguity is not a flaw to be eliminated but a structural feature that supports communicative efficiency [Piantadosi et al., 2012, Hermalin and Regier, 2019]. On this view, ambiguity is a natural consequence of the pressure to communicate efficiently about a vast and complex world with a finite, reusable set of forms. A compact vocabulary can only cover an open-ended space of meanings if forms are allowed to carry more than one sense, with context doing the work of disambiguation. Network analyses of language are consistent with this, treating ambiguity as a resource that increases expressive power while minimizing effort [Solé and Seoane, 2015].

Information-theoretic accounts further show that all efficient communication systems will be ambiguous if context carries information about meaning [Piantadosi et al., 2012]. Ambiguity enables the reuse of compact, efficient linguistic units, with context selecting the intended interpretation. Visual symbolic systems display similar properties [Cohn, 2013]: symbols in pictorial narratives or ideographic languages follow conventional compositional rules yet remain underspecified, with meaning resolved through surrounding context. The same visual form can activate distinct interpretations depending on its narrative, situational, and cultural embedding.

These literatures are resonant with the relational view present in indigenous epistemologies. From a semiotic perspective, meaning emerges through a triadic relation among sign, object, and interpretant, with the interpretant mediating how the sign is understood in a given situation [Kilstrup, 2015, Smythe and Chow, 1998, Innis, 1985]. Context helps stabilize one interpretation among the many the sign may support. Saussure's relational language and Peirce's triadic sign converge, from a Western direction, on a claim the Lakota tradition states directly, that a thing's meaning is its place among relations.

3 Modeling Contextual Salience in a Symbolic Field

Building on this notion of ambiguity, we operationalize meaning as a dynamic, context-sensitive process. Instead of a static vector, each symbol is defined by a set of textual descriptors spanning its possible interpretations (e.g., "flight," "balance," "fertility" for the dragonfly). Each descriptor is embedded with a state-of-the-art language model (Qwen3-Embedding-0.6B [Zhang et al., 2025]), giving a row-normalized descriptor matrix $D \in \mathbb{R}^{n \times d}$ for the field. This section characterizes the field *at rest* and introduces the point-level view of contextualization, namely which descriptors a context foregrounds. Section 4 then lifts this to the relational level.

3.1 The symbolic field at rest

Inspired by prior work on semantic-network analysis and distributional semantics [Steyvers and Tenenbaum, 2005, Mikolov et al., 2013, Arora et al., 2017], we quantify the structure of symbolic ambiguity in the meaning space with three complementary metrics. *Dispersion* is the average dissimilarity among a symbol's descriptor embeddings (spread of its facets); *leakage* is the proportion of each descriptor's nearest neighbors belonging to other symbols (cross-symbol overlap); and *inter-symbolic entropy* is the average Shannon entropy of the softmax-normalized similarities between each descriptor and all symbol centroids (how evenly relevance can be distributed across interpretations). Together these distinguish symbols whose ambiguity is internal (dispersion) from those whose ambiguity arises from external overlap (leakage) or context-sensitive spread of relevance (entropy). Figure 2 illustrates this structure.

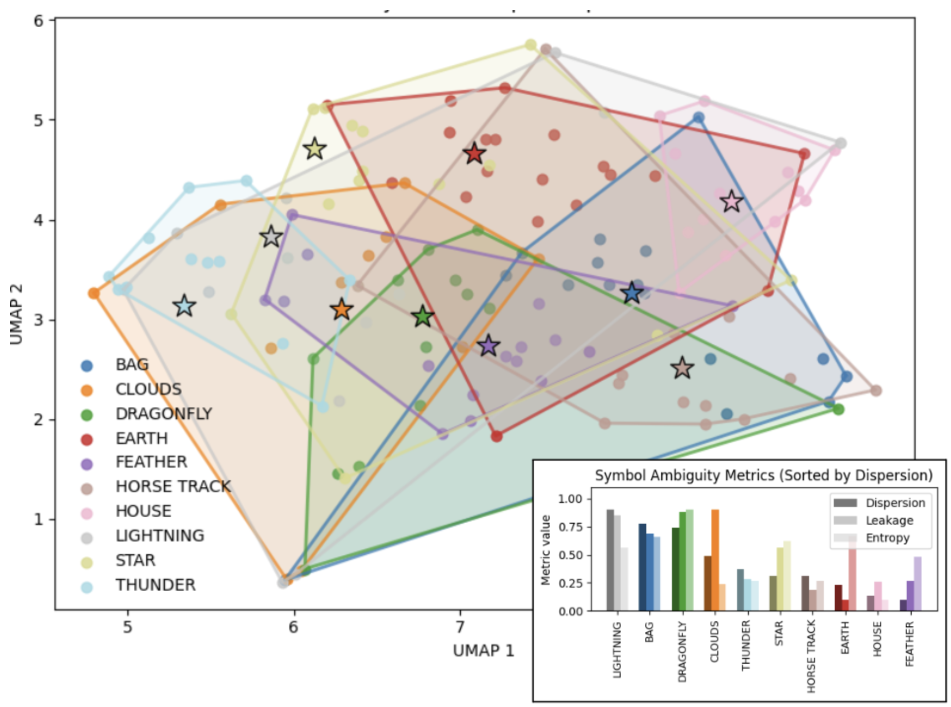

Figure 2: Symbol–descriptor semantic map with ambiguity metrics. A UMAP projection [McInnes et al., 2018] shows all symbol descriptors (points, color-coded by symbol) in shared embedding space. Stars mark symbol centroids; colored convex hulls outline each symbol's semantic extent, with overlaps indicating shared meaning. The inset bar chart reports dispersion, leakage, and entropy, sorted by dispersion. Together, the map and metrics reveal both clearly bounded symbols and those with high contextual fluidity or semantic overlap.

3.2 The contextual similarity distribution

We then use context to probe how these structural properties relate to the facets that become salient in a given situation. Context can be provided either as a weighted set of neighboring symbols (e.g., Clouds: 0.3, Earth: 0.2) or as a free-form sentence. The context vector c, encoded with the same model, induces a *contextual similarity distribution* over each symbol's descriptors. Cosine similarities between c and each descriptor are passed through a softmax, so the most contextually similar descriptors carry the most weight. Figure 3 shows this distribution for the symbol Earth across four contrasting emotional themes (Fear, Love, Sadness, Gratitude). Each theme is a set of ten short sentences written to convey that emotion (for Fear, e.g., sentences describing threat and dread), used here as contextual probes and which were chosen as distinct, contrasting affective registers to give a broad range of contextual variation. Different descriptors gain prominence under different emotional framings, a clean demonstration of context-conditioned salience.

3.3 Graph-based relational salience

Beyond the per-descriptor similarity distribution, our framework models the symbolic system as a bipartite symbol–descriptor graph, with additional descriptor–descriptor links where embeddings are highly similar. A Personalized PageRank (PPR) process [Lofgren et al., 2016] can be initialized either from symbol weights or from a free-text context, distributing personalization mass to descriptors by semantic similarity and propagating it through the graph. The novelty of this application lies in using embedding-based cosine similarity as the mechanism through which free-text context enters and guides a graph-based propagation process. After convergence, each symbol's PPR score reflects the mass it absorbs via all direct and indirect paths from the context, capturing structural salience from both semantic and topological proximity. We define a symbol's *predicted salience* flexibly,

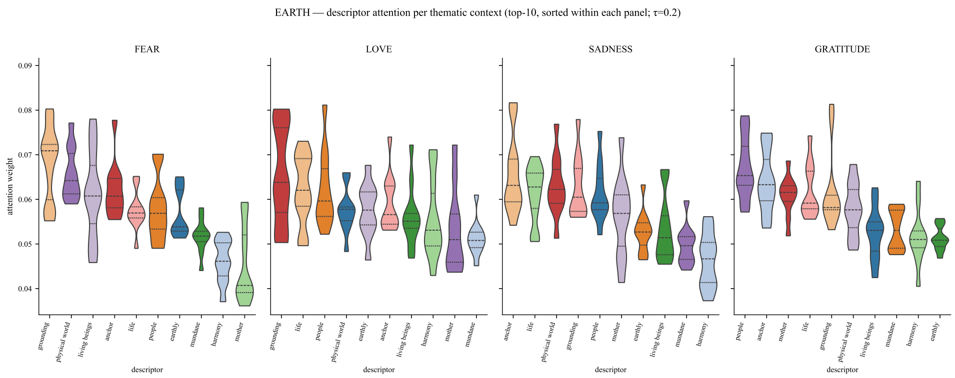

Figure 3: Contextual similarity distribution for Earth's descriptors across four emotional themes. Each theme comprises ten short sentences written to convey the corresponding emotion (Fear, Love, Sadness, Gratitude), used as contextual probes. Violin plots show, per theme, the distribution of each descriptor's weight over the ten probe sentences, sorted by median weight; higher weights indicate greater cosine similarity of a descriptor to the thematic context. (Weights are a softmax over context–descriptor cosine similarities, a similarity score, not transformer attention.)

using context-conditioned embedding similarity, PPR, or their combination to capture semantic fit within the broader structure of the network.

The contextual subgraph (Figure 4) offers an interpretable view of how a given input "activates" the symbolic network, surfacing latent thematic clusters and intermediary concepts that would be obscured in the full graph. Figure 5 complements this with a topology-level view of the same salience score. The procedure fixes a *root* symbol; for every other symbol, it builds a two-symbol context (root + partner) and computes each remaining symbol's predicted salience as defined above (embedding similarity combined with PPR); then, for each target symbol, it averages the salience increase across all partners and bootstraps the spread ($n = 20$ random partner-weightings). In each star, the *bar length* for a target symbol indicates its mean salience increase across all root+partner contexts, that is, how strongly the root tends on average to make that target more salient. The *radial violin* around that bar shows how much this effect varies across different partners and weightings. Long bars therefore mark targets that are strongly co-activated by the root on average, while wide violins mark targets whose co-activation is especially context-dependent. Together, the plots distinguish stable semantic neighbors from relationships that depend strongly on the accompanying symbol, revealing structure that embedding similarity alone would not separate.

Together, these analyses establish methods to quantify how context redistributes salience across descriptors and symbols. This is real and useful, but it operates on individual nodes. A relational epistemology asks for more. It asks how context reshapes the *relations among* descriptors.

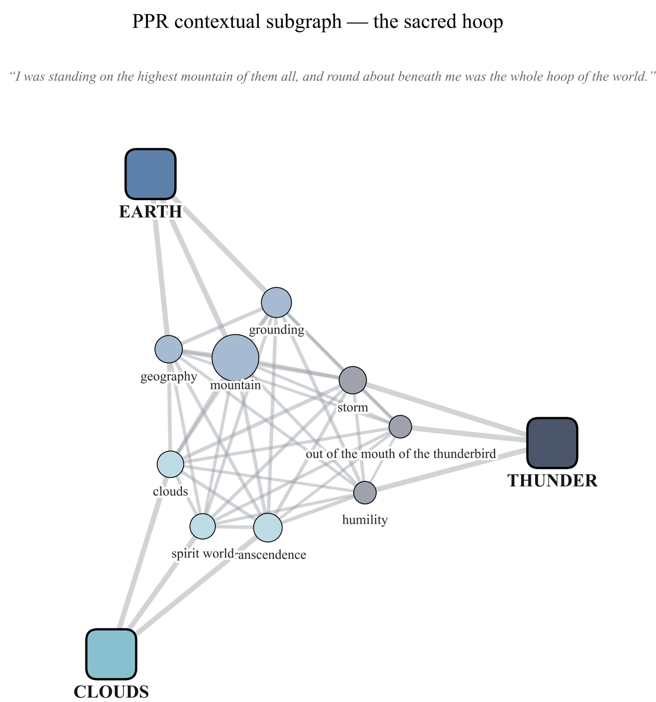

Figure 4: Contextual subgraph from Personalized PageRank for a narrative fragment. Symbols (squares) and top descriptors (circles) are colored by symbol, with edge thickness indicating association strength. Baseline centrality is subtracted to highlight relevance emerging specifically from the current context.

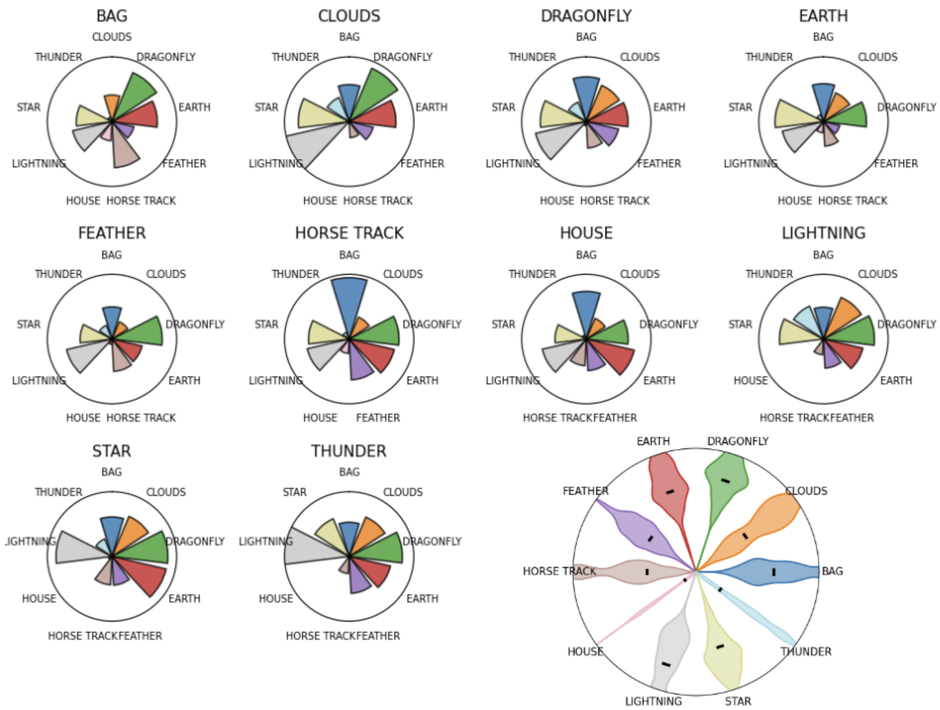

Figure 5: Star plots of pairwise contextual salience. For each *root* symbol (one panel), we form a context from the root paired with each other symbol in turn and record the predicted salience (context-conditioned embedding similarity combined with PPR score, §3.3) of every remaining symbol. Each point on a star is a target symbol; the bar is its mean salience increase over all root+partner pairings, and the radial violin summarizes the distribution of that increase across pairings (bootstrapped over randomly assigned partner weights, $n = 20$). Long bars mark targets that the root reliably makes salient regardless of partner; wide violins mark targets whose salience depends strongly on which partner is present.

4 Quantifying Contextual Recoupling of Latent Semantics

The similarity-distribution and PPR views above answer *which* descriptors a context foregrounds. They do not directly measure how a context reshapes the geometry that holds the field together, the pattern of pairwise similarities among descriptors. If meaning lives in relations, this geometry is the object of interest, and a context's effect on it is the quantity to measure.

4.1 The Δ coupling matrix

A context-shift operator S maps the descriptor matrix D to a context-shifted matrix $D' = S(D, c, \theta)$. Throughout we use a gated additive operator,

$$D' = \text{normalize}\big(D + \beta\, g_\tau(\langle D, c\rangle)\, c\big)\,, \tag{1}$$

which nudges each descriptor toward the context c in proportion to a gated relevance $g_\tau(\langle d_i, c\rangle)$ (defaults $\beta = 1.2$, $\tau = 0.3$, with per-symbol softmax gating); other strategies (re-embedding in a context-templated prompt, pooled blends, convex hybrids) fit the same slot. The latent geometry of the field at rest is the Gram matrix $C_0 = DD^\top$ of pairwise inner products; under context it becomes $C_1 = D'D'^\top$. The Δ **coupling matrix** is their signed difference,

$$\Delta = D'D'^\top - DD^\top, \qquad \Delta_{ij} = \text{change in coupling between descriptors } i \text{ and } j, \tag{2}$$

with the diagonal zeroed. Δ is a direct, per-relation measurement of the *recoupling* a context induces in the field's latent geometry, where positive entries mark descriptor pairs that context pulls together and negative entries pairs it pushes apart. Rendered as a signed graph over the strongest $|\Delta_{ij}|$ edges, or aggregated to a between-symbol drift map, it localizes contextualization to specific lexical relations rather than to point displacement.

Figure 6 shows three contexts producing three distinct recoupling patterns over the same field. This is the Shape Kit's interpretive openness made computationally visible. A single curated lexicon supports many relational configurations, selected by narrative framing.

4.2 Interpreting and calibrating the Δ readout

The raw Δ matrix answers a local question: which descriptor relations does a given context strengthen or weaken most? Figure 6 therefore displays the strongest raw edges for each context, without centering. Although the complete Δ fingerprints of C_1, C_2, C_3 are nearly collinear (pairwise cosine ≈ 0.99), reflecting a broad increase in coupling shared across contexts, their strongest edges are largely distinct (top-30 overlap 2–9%). Selecting these peaks suppresses much of the common background and makes the context-specific relational structure visible. Peak selection still leaves symbols that respond strongly across many contexts in place, and the fully context-specific signature of a context emerges once the cross-context baseline is removed.

For analyses based on the full fingerprint rather than a small set of top edges, we therefore explicitly center the readout. Given a reference set of K contexts, we compute

$$\widetilde{\Delta}_k = \Delta_k - \frac{1}{K}\sum_{\ell=1}^{K} \Delta_\ell,$$

so that $\widetilde{\Delta}_k$ retains the relations strengthened or weakened by context k relative to the average contextual response of the field. The reference set is chosen once and applied to every later context. It works best when it samples the range of framings the field will meet, broad and varied enough for its mean to capture the field's generic contextual response, and a few dozen oblique probes suffice in practice. We use thirty; the baseline is a smooth, low-variance object that stabilizes quickly as probes accumulate, so the diversity of the probes carries more weight than their number. Symbol-level responses derived from these residual couplings are subsequently z-scored across symbols. Raw Δ is thus appropriate for inspecting the strongest relations activated by one context, whereas $\widetilde{\Delta}$ supports comparisons that would otherwise be dominated by the common coupling baseline.

The structure measured by Δ is, to high accuracy, a first-order consequence of descriptor relevance and resting geometry. Let $s_i = \langle d_i, c\rangle$ denote the relevance of descriptor i to context c. Before renormalization, the gated shift yields

$$\langle d_i', d_j'\rangle \propto C_{0,ij} + \beta(g_j s_i + g_i s_j) + \beta^2 g_i g_j,$$

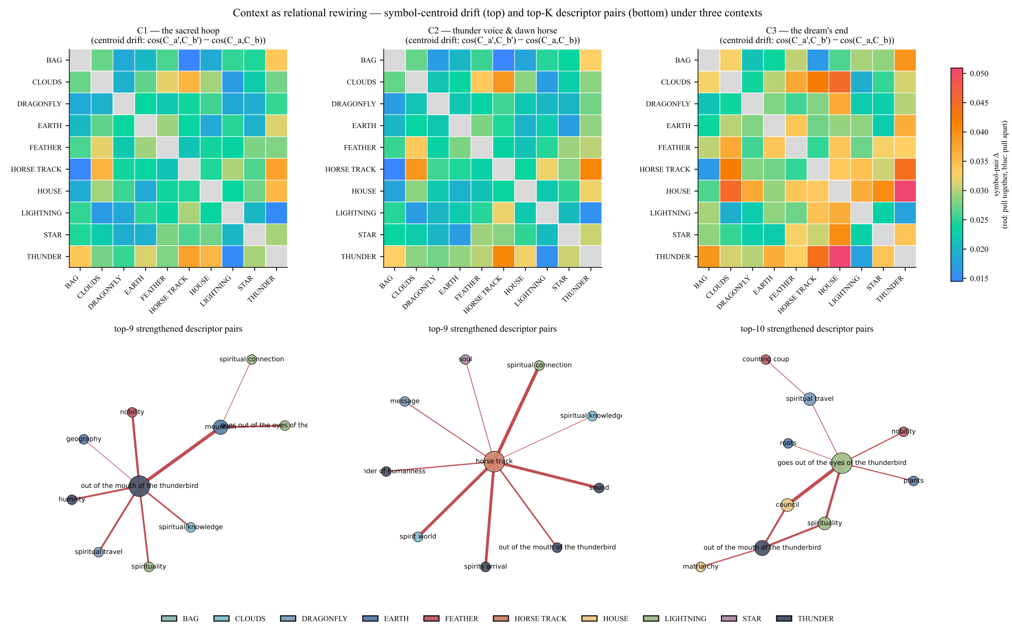

Figure 6: Context as relational rewiring. *Top:* between-symbol centroid-drift maps for three narrative contexts C_1/C_2/C_3 over the full Shape Kit ($\cos(C'_a, C'_b) - \cos(C_a, C_b)$; warm = symbols pull together, cool = pull apart). *Bottom:* the corresponding Δ-graphs of the top strengthening descriptor pairs (raw Δ, no centering). The same lexicon yields markedly different relational structures under different framings, their strongest edges near-disjoint across contexts (top-30 overlap 2–9%), with edges clustering intra-symbol under one context and bridging symbols under another. Context does not move the symbols; it activates particular relations among their descriptors.

so a pair strengthens primarily when its descriptors are context-relevant, conditional on how strongly they were already coupled. A model based only on $s_i s_j$ explains part of the observed change ($R^2 \approx 0.1$–0.5), whereas adding $C_{0,ij}$ and its interactions reconstructs the gated-operator Δ almost completely ($R^2 \approx 0.99$). The readout therefore does not recover a hidden higher-order structure unavailable to the encoder. Its contribution is to re-express the encoder's contextualization as a grounded, per-relation pattern in the coordinates of a community-curated symbolic field.

4.3 Reading a narrative as a trajectory through coupling space

Because Δ is defined per context, a narrative read step by step traces a *path* through coupling space. Figure 7 feeds a twelve-step generic story (a day on the plains; plain English, written for this figure) to a readout calibrated once on a fixed reference battery of oblique probes, and tracks three views, a symbol timeline (the de-biased per-symbol response, whose hot spot migrates with the arc, from open grass and horses to a building storm, the strike, shelter, and the night sky), the story as a connected trajectory in the principal plane of the residual recouplings, and the symbol–symbol couplings with the largest range over the story. Meaning does not sit still as a story unfolds; the recoupling readout makes its movement legible.

4.4 The recoupling margin as an ambiguity meter

The trajectory view tracks *where* the recoupling points at each step; we now ask *how decisively* it points there. The same de-biased per-symbol response that drives the timeline in Figure 7 has a top-1 and a top-2 at any given context, and their *margin* measures how sharply the context resolves the field to a single symbol. A large margin marks a settled reading; a small margin marks a genuinely open one, where two or more symbols remain in play. The margin is thus an ambiguity meter, and Figure 8 shows it behaving as one. On a battery of indirect single-target probes, short phrases designed to evoke one symbol without naming it, the margin separates the readout's settled readings from its open ones (AUC ≈ 0.78). Phrases deliberately constructed so that two symbols fit equally well fall

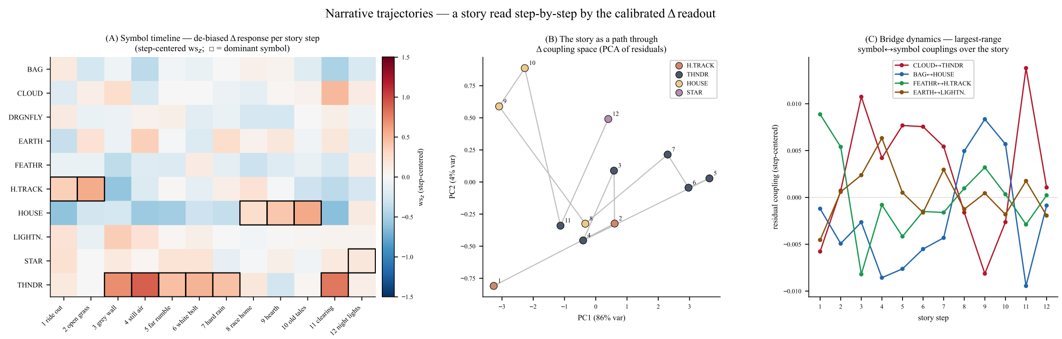

Figure 7: Narrative trajectories. A twelve-step story read by the calibrated Δ readout. *(A)* symbol timeline, the de-biased per-symbol response per step, the dominant symbol boxed, the hot spot following the arc. *(B)* the story as a connected path through coupling space (PCA of per-step residual recouplings). *(C)* dynamics of the four highest-range symbol–symbol couplings. The story is a trajectory, not a point.

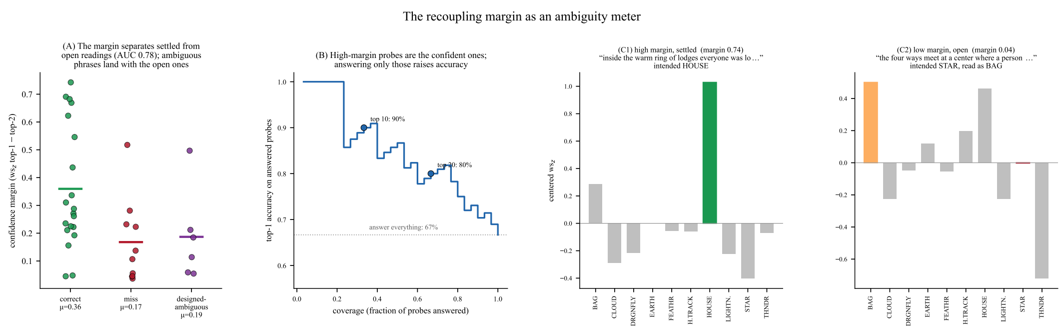

Figure 8: The margin as an ambiguity meter. *(A)* the confidence margin (top-1 minus top-2 de-biased response) separates settled readings from open ones (AUC 0.78); phrases written to be deliberately ambiguous land with the low-margin cases. *(B)* the same quantity doubles as a confidence score, and ranking probes by margin lifts top-1 agreement with the intended symbol from 67% to $\sim 90\%$ over the most confident third. *(C)* two response profiles, a peaked reading that commits to one symbol (House) beside a low-margin one whose mass is split across competing symbols (Star and Bag), the meanings a navigation interface would surface together.

in the same low-margin region. These low-margin cases are inputs the phrase leaves open, and the response profile hands back exactly which symbols are in tension (Figure 8C). This is the signal a system for navigating ambiguity needs. Where the margin is high, it can commit to one reading; where it is low, it can surface the competing interpretations rather than force a single answer.

4.5 A numerical context-to-context similarity: the field-specific kernel

The Δ-graph of §4.1 reads a single context from the inside, showing which descriptor relations one framing strengthens or weakens. This subsection asks a different question one level up, namely which of many contexts act on the field the same way. The unit of comparison is no longer a descriptor pair but a whole context. We represent each context by its entire Δ fingerprint, the vector of all its coupling changes, and define the similarity between two contexts as the cosine between their fingerprints. Arranged as a matrix over a corpus, this *context kernel* answers which framings recouple the field alike rather than which symbols are linked, so it is a different instrument from the Δ-graph rather than a summary of it. The two are complementary, the graph being the anatomy of one recoupling and the kernel a map of how many recouplings relate.

Concretely, we run each context in a fixed battery through the calibrated readout, take the upper triangle of its Δ matrix as a fingerprint, and stack the fingerprints into one matrix with a row per context. We subtract the mean fingerprint, so that what remains is how each context departs from the average framing, and take the cosine kernel of these centered fingerprints. Because the kernel sums over the *entire* fingerprint rather than a handful of top edges, it needs one precaution the subgraph reading did not, and surfacing that precaution is part of the method.

Taken in this raw form the kernel is dominated by a single mode, about 82% of the variance (Figure 9A), that captures not *how* a context reshapes the field but merely *how strongly* it engages the descriptor bank at all (correlation 0.84 with a pure activation-strength predictor). Overall activation strength is a generic property of a context rather than a fact about the Lakota field, which is why the raw kernel barely changes when the same contexts are read through an unrelated descriptor bank (a generic six-emotion set, whose raw kernel agrees with the Lakota one at $r = 0.93$). Before correction the kernel measures context "loudness" rather than the structure of the symbolic system.

One standard, label-free operation removes the mode. We take the singular value decomposition of the centered fingerprints and subtract their top principal component, the single direction that carries that 82% of the variance ("all-but-the-top" [Mu and Viswanath, 2018]). Because that direction is the loudness axis, removing it leaves each context's content, the way it recouples the field differently from the average, in the directions orthogonal to sheer magnitude. Afterward the loudness mode is gone (its correlation with activation strength falls $0.84 \to -0.13$) and the kernel becomes *field-specific and content-bearing*. It now depends on the chosen lexicon (cross-lexicon agreement drops $0.93 \to 0.76$, so the similarity reflects the Lakota structure rather than a generic magnitude), it aligns with sentence meaning (agreement with sentence-embedding similarity rises $0.31 \to 0.69$), and hierarchical clustering recovers coherent thematic registers, each readable off its own representative sentence in Figure 9B.

The cleaned kernel turns a corpus of contexts into a navigable map of *registers of framing*. Two operations make this actionable. First, clustering the kernel over a corpus groups its passages by how they engage the field, so a collection of dream narrations, or the successive steps of one story, resolves into a handful of recurring framings, each summarized by the passages that fall in it, rather than a flat list to read end to end. Second, ranking the corpus by kernel similarity to a query passage returns the passages that recouple the field the same way, a retrieval that follows effect on the symbol system rather than surface wording, so two passages worded very differently but engaging the same symbols land together. Because both the groups and the neighbours are labelled by their own sentences and legible as themes in the field's categories, a knowledge holder can read, check, and steer the result instead of trusting an opaque score.

This clustering is not the same as grouping the sentences by embedding similarity directly. The two agree only at $r = 0.69$, and unlike sentence similarity the kernel is field-specific, changing with the lexicon, so it groups contexts by their effect on this system rather than by generic meaning. The same construction reaches across symbol systems. Because every system yields an $n \times n$ kernel over one shared battery of contexts, two systems can be compared by how differently they organize the same material. A comparative study of the distinct modes of engagement different systems bring to a text, treating community-held systems under their own consent, is a natural continuation of this work.

4.6 Robustness: is it signal, and is it encoder-specific?

A natural question is whether these recouplings *matter*, whether they are structured signal rather than an inevitable consequence of moving the descriptors at all, and whether they are an artifact of one encoder. Two compute-only checks address both (Figure 10).

Null model. For each oblique probe we replace its real context-shift with a *random* shift of equal magnitude ($\|D' - D\|_F$ matched to within 4%) and ask whether the de-biased readout still recovers the intended symbol. It does not. Real contexts ground at 67% top-1 / 93% top-3, while magnitude-matched random shifts collapse to 8% / 31%, essentially the 10% / 30% chance rates. The measured recouplings are therefore structured signal tied to a context's meaning, not a by-product of the shift's size.

Multi-embedder. Re-running the grounding battery on three unrelated encoders (Qwen3-0.6B, 1024-d; MPNet, 768-d [Song et al., 2020]; and MiniLM, 384-d [Wang et al., 2020]), grounding holds throughout (top-1 67%, 70%, 57%; all far above chance), and the three encoders assign the same symbol to a probe 73% of the time. Accuracy tracks encoder capacity (the smallest model, MiniLM, is weakest, as expected), but the readout reflects the symbolic system and the probe se-

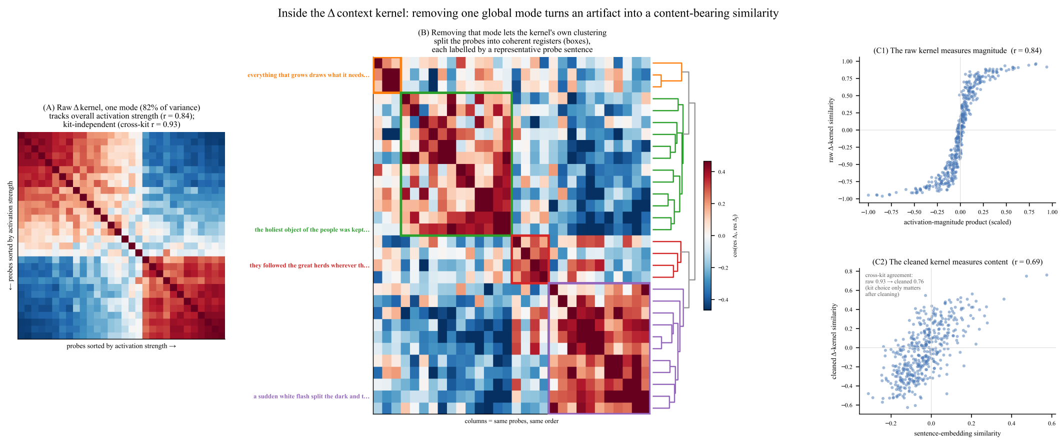

Figure 9: Inside the Δ context kernel. *(A)* the raw kernel, probes sorted by activation strength; one mode (82% of variance) tracks overall activation magnitude ($r = 0.84$) and is lexicon-independent (cross-kit $r = 0.93$). *(B)* after removing that single mode, the kernel's own hierarchical clustering (dendrogram, right) splits the probes into coherent registers (boxes), each labeled by a representative probe sentence and colored by cluster. No category labels are used; the groups are the kernel's, and the reader judges their coherence from the sentences. *(C)* the raw kernel measures magnitude; the cleaned kernel measures content (agreement with embedding similarity $0.31 \rightarrow 0.69$).

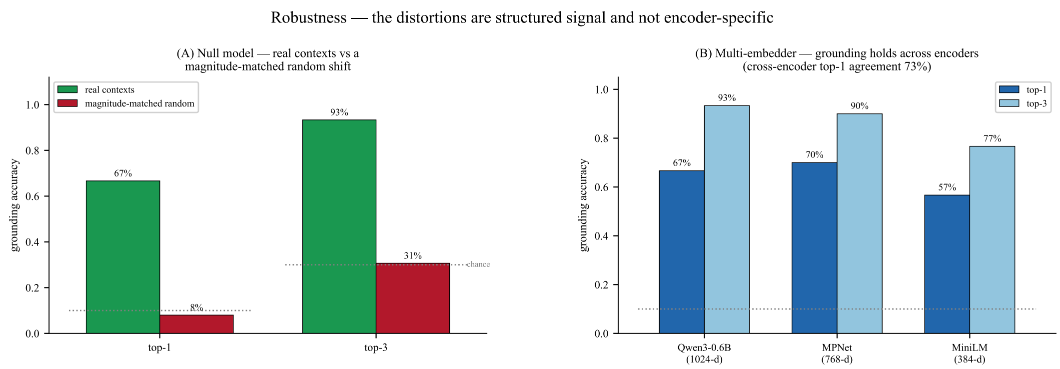

Figure 10: Robustness. *(A)* Null model. Real contexts recover the intended symbol far above a magnitude-matched random shift (top-1 67% vs 8%; top-3 93% vs 31%; chance 10/30%), so the recouplings are structured signal, not a by-product of shift magnitude. *(B)* Multi-embedder. Grounding holds across three unrelated encoders (Qwen3, MPNet, MiniLM), with 73% cross-encoder agreement on the assigned symbol, showing the readout is not an artifact of one model's geometry.

mantics rather than any single model's geometry. These checks answer the "signal vs. noise" and "one-model artifact" objections; validating that the recouplings constitute meaningful readings of a symbolic system.

5 Probing Symbolic Ambiguity Across Modalities

The Δ readout depends only on a context *vector* in the descriptor space; nothing about it is specific to text. Any encoder that lands an input in that space can drive it, so the ambiguity of a curated symbolic field can be probed not only through written context but through other modalities, fitting for a tradition carried by story, song, images and sound as much as by writing. We demonstrate this with audio, treating sound as a first-class probe of the symbolic field.

We swap the text encoder for a contrastive audio–text encoder (CLAP [Wu et al., 2023]), re-encode the field's descriptors with its text tower, and drive the readout with a recording taken straight from its audio tower. Because CLAP is trained to align sound with acoustic language, real recordings ground

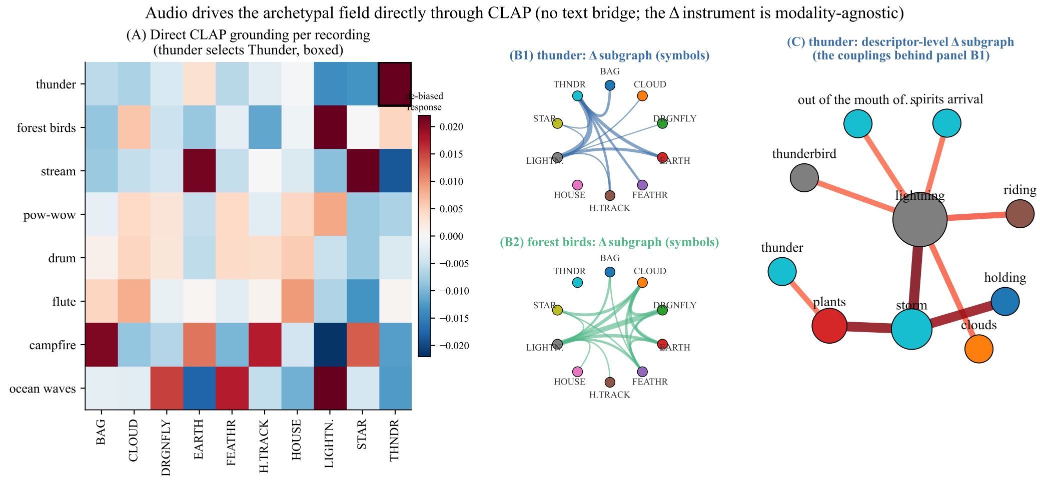

Figure 11: Audio drives the archetypal field directly through CLAP, with no text bridge. The Lakota descriptors are re-encoded with CLAP's text tower and each recording is the context straight from CLAP's audio tower. *(A)* de-biased symbol response per recording; a thunderclap selects Thunder cleanly (boxed), while recordings with no correspondent in the field (campfire, ocean) respond more loosely. *(B)* the symbol-level Δ subgraph each recording induces, ribbon width the strengthened between-symbol coupling, thunder concentrating on Thunder and Lightning and forest birds on a different set. *(C)* the descriptor-level Δ subgraph behind thunder's chord, with *lightning* the hub linking *thunder*, *storm*, *thunderbird* and *clouds*. Because CLAP aligns audio with acoustic language, the same Δ instrument reads sound as it reads text.

into the field with no intermediate bridge (Figure 11). A thunderclap most strengthens the descriptors *thunder*, *thunderbird*, *lightning* and *storm* and selects Thunder as its symbol (Figure 11A), and each recording induces its own Δ subgraph, read at the symbol level (Figure 11B) and down to individual descriptor couplings (Figure 11C). The same instrument that reads a sentence can read a sound.

This approach opens a variety of creative uses. A sound need not name a symbol to surface its meaning, and a recording whose acoustics sit between registers pulls several symbols at once, ambiguity made audible, just as oblique text did in §4. Sound is thus a second, independent window onto the same computational account of symbolic ambiguity.

6 Creative Applications

The same readouts can function not only as analytical instruments but as compositional ones. The multimodal probe of §5 offers a first creative affordance: a *sonic interface to the symbolic field*. A performer or installation can drive the field through sound in real time, while transitions between recordings trace continuous paths through the space between interpretations, supporting improvisatory storytelling that keeps a sound-carried tradition in sound. The framework also supports *associative expansion*: the Δ-graph and contextual subgraph of a narrative fragment reveal strengthened relations, symbolic bridges, and latent pathways that may extend what a poet, storyteller, musician, or visual artist might reach for [Varshney et al., 2016, Gross et al., 2012, Rick et al., 2023]. These structures can in turn become *generative constraints*. Rather than conditioning a model only on retrieved passages or textual instructions, one can prompt it with the field's active relations, such as its strongest Δ-edges, the migration of a descriptor between symbols, or a turning point in a narrative trajectory. Interpolating between contexts further produces a continuous sequence of relational states, providing a controllable path between readings rather than a discrete switch. The same dynamics can guide *nonlinear story navigation*: the current relational state can open, close, or weight possible narrative branches, allowing a story to remain exploratory in low-margin regions and to converge as the field settles into a clearer configuration. More broadly, the recoupling margin makes ambiguity itself available as an artistic parameter: low-margin states can sustain several interpretations simultaneously, whereas high-margin states can draw a composition toward a more resolved symbolic configuration. The framework thus enables AI-assisted creation to operate through the relational structure of a symbolic tradition rather than merely borrowing its vocabulary or imagery.

These affordances come together in *delyrism*, an interactive web application that opens the readouts for hands-on exploration. A user drives the field with a phrase or a recording and watches the Δ-graph and the contextual subgraph update live, sees which descriptors migrate to new symbols, morphs between framings along a continuous dial, and reads every result in the field's own descriptors and symbols. The application turns the instrument into a studio where a storyteller or designer works directly with the relational structure a text or sound activates. The system can be explored at delyrism.kairos-hive.org

Beyond any single tool, the framework points toward a larger aim, design practice grounded in living symbolic traditions and accountable to the communities that hold them. The same instrument that reads how a story recouples a field runs inward as well, surfacing the structure a community's stories activate, combining the structures that several narratives induce, and offering the result to designers and knowledge holders as material for new work. Working across narratives this way raises real questions of whose stories, under what consent, and who authors and benefits from what is made, questions that call for dedicated participatory research under the community-governed commitments stated below (§Ethical Considerations). We flag it here as the affordance the framework most naturally grows toward, a generative, culturally anchored complement to the analytic instrument developed above.

7 Discussion

This paper presented the Δ coupling matrix, an instrument that reads how a context recouples the relations within a community-curated symbolic field, and put it to work on the Lakota Shape Kit across narrative, ambiguity, context comparison, robustness, and sound. The figures of Section 4 offer several views of one object, the context-induced recoupling Δ of the field's latent geometry. Figure 6 shows its *structure*, the relations a single context strengthens; Figure 7 shows its *motion*, the path that structure traces as a narrative unfolds; Figure 8 shows its *ambiguity*, how sharply a context resolves the field or holds several symbols in play; Figure 9 shows its *comparability*, how the recouplings of two contexts relate; and Section 5 shows its *reach*, the same object driven by sound as readily as by text. One measured quantity, examined at the level of a single relation, a single context, a sequence of contexts, pairs of contexts, and a non-textual input, ties these results into a single account. A shared calibration binds them further, since the timeline and the margin read the same de-biased per-symbol response, and the trajectory and the kernel operate on the same mean-removed residual, so what the kernel analysis reveals about that de-biasing carries through every view.

Read as a whole, the framework supplies a legible, relational, and community-grounded readout of how a language model represents and reshapes a curated symbolic system, and it sits at the meeting point of three programs. For computational linguistics and interpretability, it turns the opaque geometry of contextualization into a per-relation pattern in a designed coordinate system, complementing point-level probes with a view of the relations themselves. For cross-cultural knowledge representation, it lets an Indigenous symbolic tradition serve as the very coordinate system through which a model is read, so the analysis speaks in the categories that matter to the community. For creative AI, it exposes the field's ambiguity and relational pathways as control surfaces an artist can navigate. The common thread is legibility grounded in a designed field, where every reading returns to descriptors and symbols a knowledge holder recognizes.

Several avenues extend naturally from here. *An atlas of symbolic systems.* Because every curated field yields a kernel over one shared battery of contexts, many systems, from chakras and planetary symbols to Jungian archetypes, become comparable by how each engages the same text, mapping the distinct modes of engagement a passage draws from each, with community-held systems entering only under their own consent. *Tradition-anchored generative design.* Run inward, the instrument surfaces the structure a community's stories activate and offers it to designers and knowledge holders as material for new work, the direction sketched in §Creative Applications. *Further modalities.* Because the readout depends only on a context vector, images and gesture join sound and text as ways to probe the field, widening the sensory surface of the instrument. *Broader validation.* Additional lexicons and a wider encoder pool will separate the structure intrinsic to this symbolic system from the structure common to embedding models, sharpening the account this paper begins. *A shared review protocol.* Formalizing structured interpretation with knowledge holders into a repeatable protocol will let the instrument serve many traditions on their own terms.

Two commitments frame these directions. The framework reads how a model *represents* a curated field, and validating that representation against the living tradition rests with community-grounded review, which the broader project treats as its primary evaluation. The through-line returns to where the work began. If meaning is the position of a thing in a living web of relations, then the quantity to measure under context is the reconfiguration of that web. The Δ readout makes that reconfiguration legible, a formal echo of *Mitákuye Oyás'iŋ*, meaning read as relation, and change in meaning as change in relation.

8 Ethical Considerations

This project is conducted in active collaboration with Indigenous community partners, with the last author being an Oglala Lakota knowledge holder. All symbolic material and interpretations are developed in accordance with community protocols and with guidance from cultural experts; the descriptor sets and context sentences shown here are chosen for public demonstration, and materials that belong to the community are not disclosed. The work is supported by the Abundant Intelligences program, which fosters AI systems developed in respectful interaction with Indigenous knowledge systems [Lewis et al., 2020]. Our design, data use, and dissemination follow the CARE Principles for Indigenous Data Governance [Carroll et al., 2020] and the broader commitments of Indigenous data sovereignty [Kukutai and Taylor, 2016], honoring community authority over the knowledge represented here. We stress that descriptor curation by knowledge holders is not a limitation of the method but its precondition. Indigenous expertise is what gives a community-grounded system its substance, and the framework is built so that this expertise governs what the instruments read.

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